

The Jinchuan Ultramafic Intrusion: Cumulate of a High-Mg Basaltic Magma

by GANG CHAI AND ANTHONY J. NALDRETT

Department of Geology, University of Toronto, Toronto, Ontario, Canada, M5S 3B1

(Received 30 October 1990; revised typescript accepted 22 April 1991)

ABSTRACT

The Jinchuan intrusion, situated in Gansu province, China, is an ultramafic dyke-like body emplaced in the Longshoushan uplifted terrain on the southwest margin of the Sino-Korea platform. The intrusion is 6 km long, 350 m wide and hosts a major Ni–Cu sulfide deposit. It comprises three subchambers: the west, west-central, and east. The two western subchambers are narrow and deep, and both are laterally zoned from dunite in the core through lherzolite to olivine pyroxenite toward the margins. The eastern subchamber is shallow and wide, and it shows vertical stratification grading from dunite at the base upward into lherzolite and plagioclase lherzolite, then back to lherzolite at the top.

Sixty rock samples from the Jinchuan intrusion have been analyzed for major and trace elements, and 54 samples were also analyzed for the REE. All samples contain >24 wt.% MgO, with the majority having >35% when recalculated to 100% anhydrous. Negative linear correlations are observed between MgO and most other constituents (except for a few such as Na₂O, K₂O, Sr, and Rb, which may have been affected by alteration), and it appears that the rocks were essentially formed as mixtures of cumulus olivine and primary magma.

Electron microprobe analyses show olivine compositions from Fo₇₉ to Fo₈₆, with most between Fo₈₃ and Fo₈₅. The MgO/(MgO + FeO) value of the primary magma is calculated to have been ~0.64, and its MgO content is estimated to have been ~12 wt.%. Thus, the Jinchuan igneous body is probably the ultramafic cumulate portion of an intrusion of a high-magnesium basaltic magma related to continental rifting.

We suggest that the two western subchambers of the Jinchuan intrusion represent the main conduit to the original magma chamber and that their zoning was formed by flow differentiation. The eastern subchamber probably represents a higher level of the magma chamber, where crystallization was marked by convection and periodic replenishment. After consolidation, the Jinchuan intrusion was tilted to the east so the deeper parts of the western subchambers are now exposed to the same erosion level as the shallower part of the eastern subchamber.

INTRODUCTION

Parental magma composition is generally accepted as being critical to the nature of the Ni–Cu sulfide mineralization associated with mafic–ultramafic magmas (Naldrett, 1981; Ross & Travis, 1981). Ultramafic magmas generally give rise to Ni-dominated sulfides with Ni/Cu values >7 (Marston *et al.*, 1981) as in the Ni sulfide deposits associated with volcanic peridotites at Kambalda, Western Australia (Ni/Cu = 13; Ross & Hopkins, 1975; Leshner *et al.*, 1984) and the Abitibi greenstone belt, Canada (Ni/Cu > 30; Green & Naldrett, 1981), and with the dunite–peridotite intrusions at Agnew, Australia (Ni/Cu = 40; Hill *et al.*, 1985; Barnes *et al.*, 1988b) and at Dumont, Canada (Ni/Cu > 20; Eckstrand, 1975; Duke, 1980). Mafic magmas, on the other hand, commonly produce Ni sulfides rich in Cu and platinum group elements (PGE) with Ni/Cu ratios <2. Examples are the sulfide deposits at Sudbury, Canada (Ni/Cu = 1; Pye *et al.*, 1984), Noril'sk, USSR (Ni/Cu = 0.5–1; Genkin *et al.*, 1981; Grinenko, 1985) and at Duluth, USA (Ni/Cu = 0.3; Mainwaring & Naldrett, 1977; Rao & Ripley, 1983).

The Jinchuan Ni–Cu deposit in Northwest China is one of the largest magmatic deposits in the world (Wang, 1980; Chai & Naldrett, 1989), and an important feature of the sulfide ore is that it is high in Cu and PGE as well as Ni. The average sulfide ore has an Ni/Cu value of ~ 1.6 (Jia, 1986) and contains ~ 1 ppm total PGE and Au (Jia, 1986; Sun, 1986; Yang, 1989). However, the intrusive rocks that host the sulfide deposits are entirely ultramafic, ranging from dunite to olivine pyroxenite, and they have been interpreted as having formed from an ultramafic magma with an MgO content of ~ 30 wt.% (S.G.U., 1984). But because this possibility is contrary to the general relationships described above for other deposits, several questions can be posed for the intrusion and its hosted sulfide ores: (1) Can an ultramafic magma give rise to Ni sulfides rich in Cu and PGE? (2) Have some postmagmatic processes enriched the Jinchuan Ni sulfide ores in Cu and PGE? (3) Might the ultramafic rocks that host the Jinchuan deposit actually be only a part of a mafic intrusion?

Our objectives in this study were to document and describe in detail the distribution and petrography of the Jinchuan intrusive rocks, to characterize their geochemistry and define the nature of the primary magma, and to interpret the history of the intrusion.

REGIONAL GEOLOGICAL SETTING

The Jinchuan intrusion is located on the southwest margin of the Sino-Korea platform (Fig. 1), a triangular continental mass consolidated at the end of Early Proterozoic (at ~ 1900 – 1800 Ma; Li & Cong, 1980; Miyashiro, 1981; Yang *et al.*, 1986). Early Proterozoic rocks in the southwestern part of the platform are called the Longshoushan Group (Pan, 1986). The Jinchuan region is a fault-bounded, northwest-striking, uplifted terrain in which the strike of structural lineation changes in an easterly direction from east-southeast to east (Fig. 2). Shallow basins filled with Mesozoic and Cenozoic continental sediments flank both sides of the uplifted terrain, and to the south is the Qilianshan Caledonian fold belt, which embodies a huge thickness of Cambro-Silurian clastic, carbonate, and volcanic rocks. A blueschist and ophiolite belt occurs on the southern flank of the fold belt, and it has been suggested that ancient oceanic crust to the south was subducted northward beneath the Sino-Korea platform (Zhang *et al.*, 1984; Yang *et al.*, 1986). In this scheme, the Jinchuan region may represent part of the ancient continent.

The main rock types in the Jinchuan uplifted terrain are the highly metamorphosed Early Proterozoic Longshoushan Group and a less extensively metamorphosed series of Middle and Upper Proterozoic rocks. These units form a NW-trending, SW-dipping monocline with inclination of $\sim 50^\circ$.

The Longshoushan Group occurs in the northern part of the uplifted terrain and comprises the Baijiazhuizi and the Tamazhigou formations. The older Baijiazhuizi formation is believed by some geologists to be at least partly Archean in age (Jia, 1986); it consists mainly of migmatite, gneiss, and serpentine marble and is > 3500 m thick. At its northern margin, it is in contact with a major regional fault (denoted F_1), which is also the northern boundary of the uplifted terrain. Along this fault, the Baijiazhuizi formation has been uplifted > 4000 m, and it is now in contact with post-Palaeozoic strata to the north (S.G.U., 1984). The younger Tamazhigou formation consists of various gneisses, schists, and banded marbles. The age of this formation must be greater than that of a light gray pegmatitic granite that intrudes the Tamazhigou formation and that has yielded a K–Ar isotopic age of 1719 Ma (Jia, 1986).

The southern part of the uplifted terrain features upper Precambrian supracrustal rocks. They strike NW–WNW and unconformably overlie the lower Precambrian strata. The

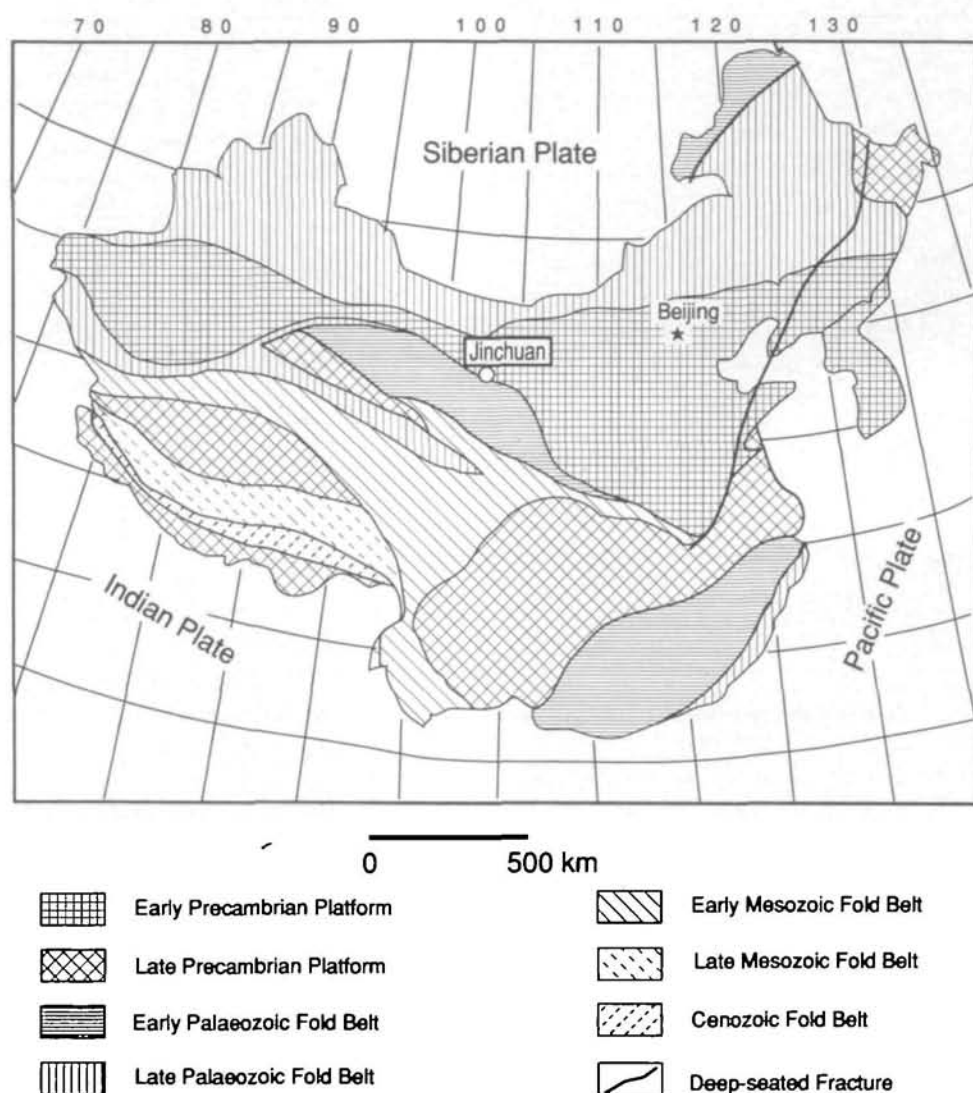


FIG 1. Map showing the location of the Jinchuan region in relation to the main tectonic units of China (modified from Yang *et al.* (1986)).

lower unit, known as the Denzhigou formation, consists of conglomerate, sandstone, and crystalline limestone. The upper units, called the Haimoshan formation, are composed of sericitic quartz schist, carbonate schist, and limestone, and are intercalated with andesitic volcanic rocks to the west.

A typical feature of the uplifted terrain is the development of different types of faults. Those forming the boundaries of the terrain (denoted F_1 and F_2 in Fig. 2) strike $\sim N50^\circ W$ in the west, but their strike changes eastward to WNW–ESE before they are lost completely under the desert sands. A series of secondary faults parallels the main faults, and they increase in number toward the main faults, forming two parallel zones. Faults F_1 and F_2 may be different in age: F_1 fault cuts through the Longshoushan Group and has

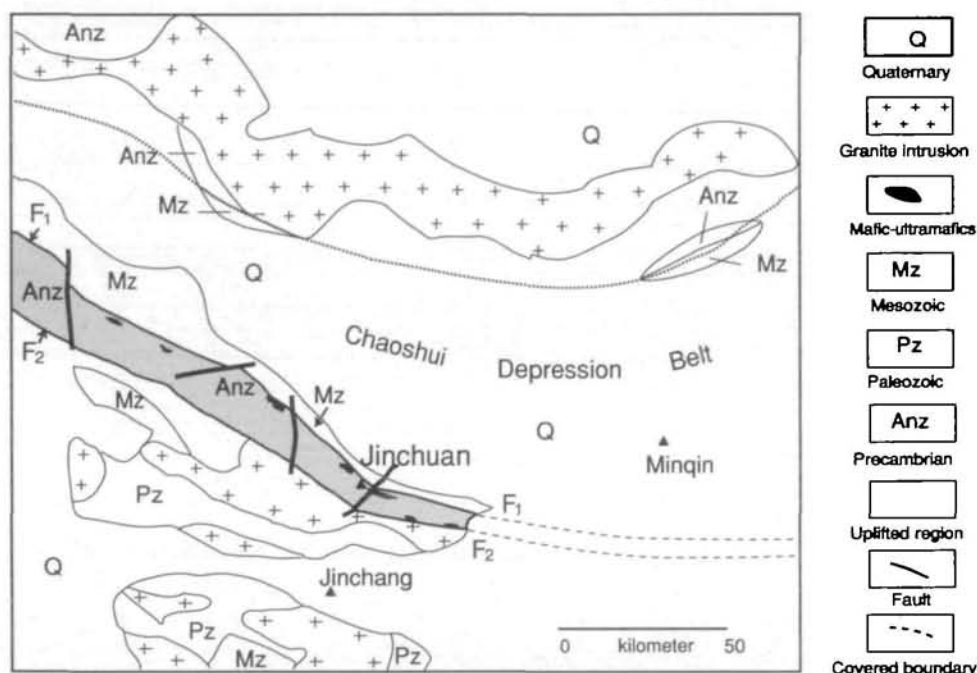


FIG. 2. Geology of Jinchuan region (compiled from the Jinchuan mine information, and from Fan (1988)).

mafic-ultramafic intrusions along it that are all early to middle Proterozoic in age (S.G.U., 1984); hence it may itself be Early-Middle Proterozoic. By contrast, F_2 cuts Middle and Upper Proterozoic rocks, and has mafic-ultramafic intrusions along it that are possibly Palaeozoic; thus it may be much younger.

Another group of faults, primarily strike-slip faults, trends northeast. A few of them cut across the whole terrain, but most occur only within it. These faults are younger than the SE-striking faults and have offset them to some extent.

GEOLOGY OF THE INTRUSION

The Jinchuan intrusion is the largest mafic-ultramafic intrusion known in the region. It is located in the northern edge of the uplifted terrain, and it cuts marbles and gneisses of the Baijiazhuhi formation (Fig. 3.1). The intrusion has a dyke-like form and generally strikes NW-SE, roughly parallel to the regional structural strain. It is ~6000 m long, 300 m wide, and >1000 m deep in its central part. It is relatively well exposed, except at its eastern and western extremities, but the rocks are heavily oxidized and appear rubbly on outcrop.

The intrusion is intersected by a series of northeast-trending strike-slip faults, and because of them, there has been a sequential discovery of the sulfide ore bodies within different sections. In accordance with the exploration order of these sections, the intrusion and its sulfide ores have been divided into four mining areas, known formally, from west to east, as Mining Areas III, I, II and IV (S.G.U., 1984).

Mining Area III covers the western 600-m segment of the intrusion where it has been offset 900 m to the southwest and isolated by fault F_8 . The segment is wedge-shaped and widens to

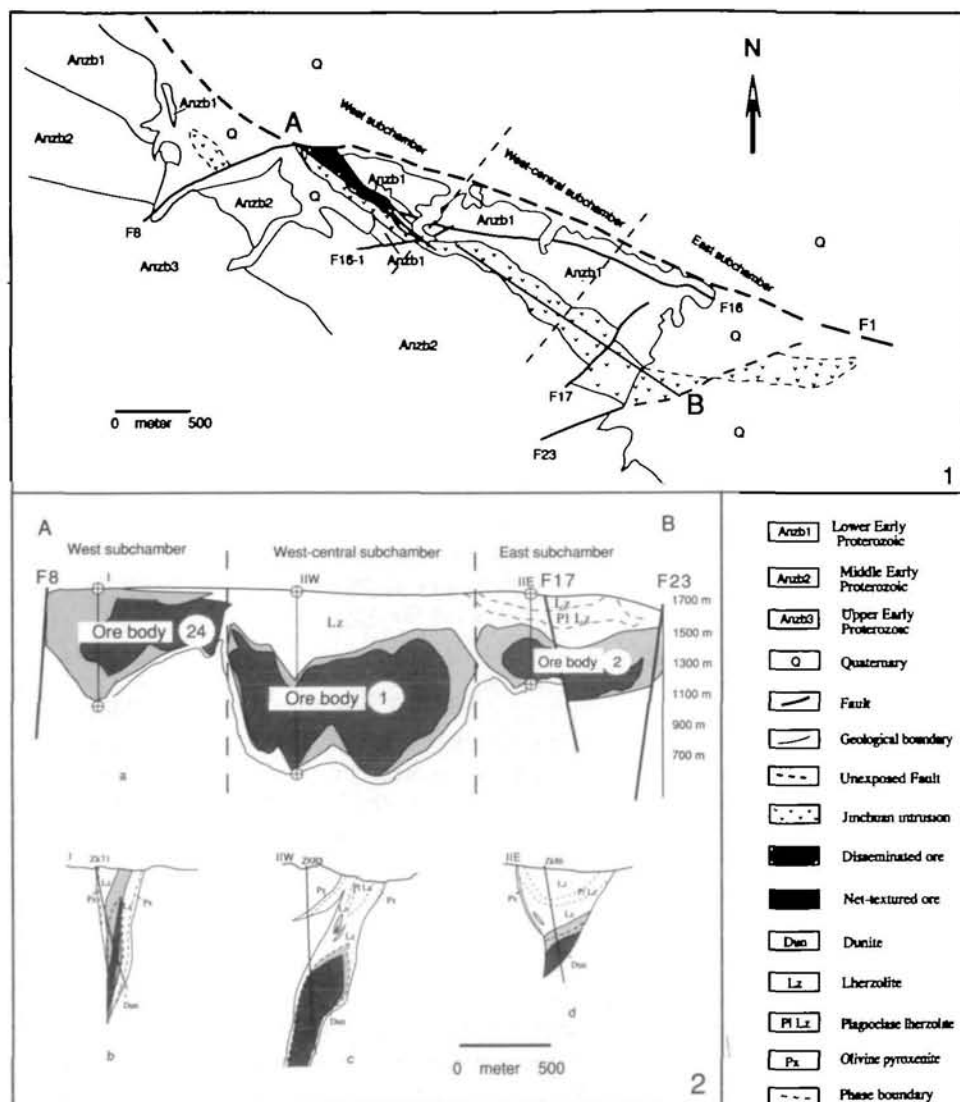


FIG. 3. Geology of the Jinchuan intrusion. 1. Surface geology [modified from Jia (1986)]. 2. Longitudinal and cross-sections [from S.G.U. (1984)]. (a) Longitudinal section; (b) cross-section of west subchamber; (c) cross-section of west-central subchamber; (d) cross-section of east subchamber.

the east, reaching a thickness of 250 m at its eastern termination. This area is covered with >40 m of Quaternary alluvium.

Mining Area I is between faults F_8 and F_{16-1} . Here, the intrusion is ~1500 m long, and it decreases gradually in width from ~300 m in the west to only 20–30 m in the east. The depth of its base is ~500 m at its western end and increases gradually eastward to ~800 m over a distance of some 500 m (Fig. 3.2a,b); then it decreases to only ~200 m at the eastern end.

Mining Area II covers the longest part of the intrusion, a >3000-m segment that is narrower in the west than in the east. This segment appears continuous on the surface but actually has two parts with different features. The western part begins from fault F_{16-1} and

extends ~1700 m to the east. It is narrow and tabular, ranging in width from ~30 m at its western end to ~300 m in the east. Its base is ~500 m deep in the west, but it plunges rapidly to >1000 m in a distance of only 300 m and remains at this level for ~1000 m before it rises to ~500-m depth in the east. This segment also has an extra branch that starts from near surface and dies out at ~300–400 m. This branch hosts the largest ore body of the Jinchuan mine (Fig. 3.2a,c).

The eastern part of the intrusion in Mining Area II is shallower and wider, with a trumpet- or Y-shaped cross-section (Fig. 3.2a, d). Its maximum width is ~500 m at the surface, and its base plunges easterly from ~600 m to 800 m. Sulfide mineralization is located mainly at the base of this section, forming a lens-like mass that dips to the southwest. A northeast-trending slip fault, F_{17} , cuts off an eastern block and offsets it downward by 90–150 m, yielding a surface width of 530 m (Fig. 3.1).

Mining Area IV encompasses the eastern end of the intrusion and is separated from Mining Area II by fault F_{23} . This segment is 1300 m long, 450 m wide, and 700 m deep in the west, and becomes narrow and shallow to the east. It has been offset by fault F_{23} to the northeast, but it is still connected to the main intrusive body. The bedrock in this area is overlain by 60–140 m of Quaternary alluvium.

Although the above subdivision of the intrusion meets the needs of exploration and mining operation, it does not have much petrological significance. For example, the two ends of the intrusion should obviously be parts of their neighbouring sections in terms of geometrical and petrological continuation, but they have been subdivided into two extra areas; and Mining Area II actually consists of two petrologically different parts.

To study systematically the petrology of the Jinchuan intrusion, a petrological subdivision is necessary. On the basis of the geometry and distribution of rock types, we have divided the intrusion into three magma subchambers: the west, west-central, and east subchambers (Fig. 3.1). The west subchamber is west of F_{16-1} and comprises Mining Areas III and I. The west-central subchamber embodies the western part of Mining Area II, where the intrusion is narrowest and deepest with **the most typical lateral zonation**. The east subchamber comprises Mining Area IV and the eastern part of Mining Area II and shows a **vertical stratification of rock types**. In addition to their similarities of shape, the two western subchambers are also petrologically very similar, so they can be considered together in the following sections, where the petrology of the Jinchuan intrusion is described and discussed simply in terms of its western and eastern parts.

The age of the Jinchuan intrusion is not well defined. It is cut by numerous lamprophyre dykes, one of which has yielded a K–Ar age of 1339 Ma (S.G.U., 1984), so that is presumably a minimum age. K–Ar isotopic ages of 1505 and 1529 Ma have also been reported for biotite from the intrusion itself (Jia, 1986), but the biotite is not a primary mineral, and the intrusive rocks are intensely altered, so these numbers must be regarded with caution. Thus, the best that can be said at the moment is that the Jinchuan intrusion was probably emplaced at some time between 1719 and 1339 Ma.

PETROLOGY AND PETROGRAPHY

The Jinchuan intrusion has been divided into four main rock types on the basis of field mapping, drill-core logging and microscopic observations. They are dunite, lherzolite, plagioclase lherzolite, and olivine pyroxenite. Contacts between different rock types are gradational, and no clear divisions can be seen in the field.

The distributions of the four rock types are very different in the western and eastern parts of the intrusion. **The western part, which is narrower and deeper, features a dunite core,**

herzolite wings, and olivine pyroxenite margins. The dunite core is **lensoid or tabular** in shape, and it parallels the walls of the intrusion but is commonly closer to the footwall side. It is ~50–150 m thick and is enveloped by herzolite at most places. In the deeper part of the intrusion, the lowest part of dunite is almost entirely sulfide-bearing, and the marginal herzolite zones are very thin or absent. Herzolite is predominant at higher levels, however, reaching widths of almost 300 m at the surface, but downward in the section, it splits around the dunite to form two flanking zones. Locally at the surface, small amounts of plagioclase herzolite are encountered in narrow bands, generally < 10 m thick, but they die out quickly along strike. To the east, however, these bands occur more frequently, and eventually some of them grade into thick layers that may be the western extension of the plagioclase herzolite layer in the eastern part of the intrusion. Olivine pyroxenite invariably occurs in the margins of the intrusion, usually in a zone < 40 m thick, and it is discontinuous in some places where the intrusion narrows. In the hanging wall side, olivine pyroxenite is generally slightly thicker than in the footwall side, but the reverse relationship can also be seen at some localities. At the contact between the intrusion and the country rock, the intrusive rocks are often altered and sheared to a chlorite–serpentine schist zone up to 5 m thick.

The spatial differentiation of the rock types, from dunite in the core, through herzolite to olivine pyroxenite on the margins, is clearly concordant with the variations of MgO, SiO₂, Al₂O₃, and TiO₂ across the intrusion. Thus in the sample section in Fig. 4a, MgO increases from the hanging wall contact to the centre of the intrusion, and then decreases to the footwall contact, in parallel with the modal variations of olivine; SiO₂, Al₂O₃, and TiO₂

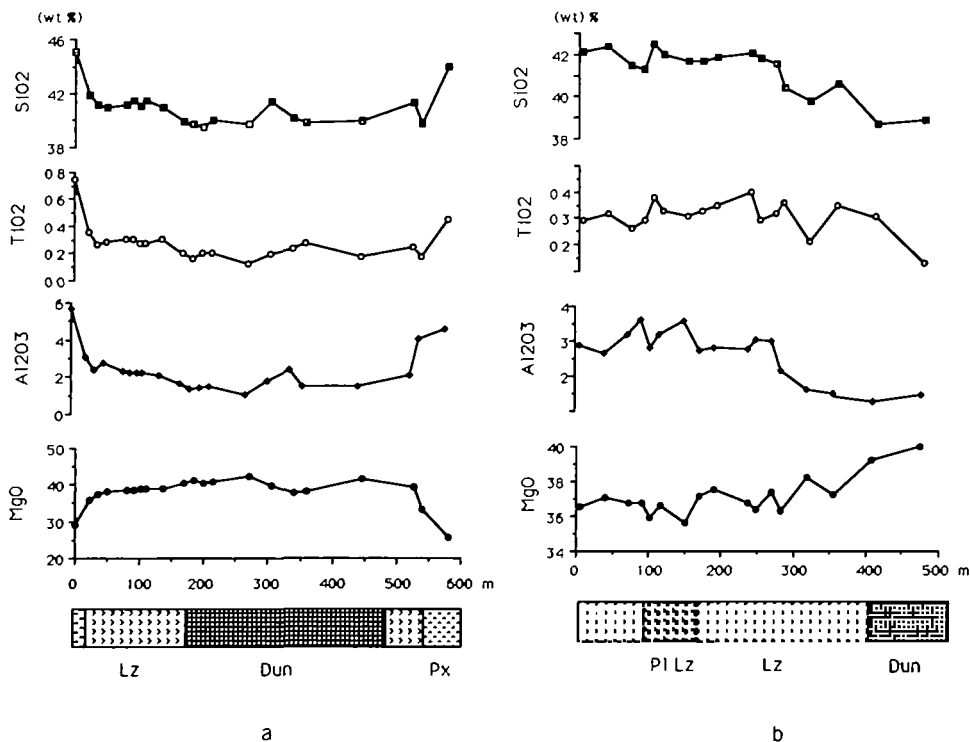


FIG. 4. Whole-rock variations of selected major elements along drill holes: (a) zk71; (b) zk86. Dun, dunite; LZ, herzolite; Pl LZ, plagioclase herzolite.

display the reverse trends. The MgO-rich section extends for ~500 m along the drill hole (not a true thickness) and is relatively uniform, with MgO generally in the range 36–41 wt.% (Fig. 4a). Vertically, in this part of the intrusion, there is generally an upward change from dunite to lherzolite with an irregular transition between them, but dunite reaches the surface in the west.

In previous studies, the lateral zoning has been emphasized, and the intrusion has been interpreted as a concentric intrusive body (Jia, 1986), similar to Alaskan-type intrusions (Irvine, 1974). However, in this investigation, we have found that the eastern part of the intrusion is distinctly different. Although the lateral variation of the rock types from dunite through lherzolite to olivine pyroxenite still exists in the east, the lherzolite is much thicker and makes up >80% of the cross-section. Dunite is largely restricted to the base of the subchamber and is closely associated with the sulfide mineralization (Fig. 3.2d); vertically, there is a rapid upward change to lherzolite and plagioclase lherzolite. Olivine pyroxenite is also less common than in the west, occurring only as marginal zones <20 m thick.

On the other hand, vertical differentiation is prominent in the east subchamber, as shown in Fig. 3.2d. There is an upward succession from dunite at the base of the intrusion through lherzolite to plagioclase lherzolite, and then back to lherzolite at the top. The cross-section in Fig. 3.2d also shows a rough layering with a saddle at the centre, in which plagioclase lherzolite becomes an important marker unit. Figure 4b shows the variations of MgO, SiO₂, Al₂O₃, and TiO₂ with depth in the eastern subchamber. It is seen that MgO generally decreases from 40 wt.% at the base to 36 wt.% at ~290-m depth and then increases slightly to ~37.5 wt.% at ~180 m depth where the plagioclase lherzolite layer begins. This layer extends to ~100 m depth, and its MgO content is only 35–36%. Above it, there is a return to lherzolite, and MgO increases to almost 37 wt.%.

Dunite in the Jinchuan intrusion is generally sulfide-bearing, with 8–30% Ni–Cu sulfide, and it hosts almost all of the net-textured mineralization of the ore deposit. Olivine, of course, is the dominant mineral; pyroxenes, chromite, and in places, plagioclase are minor phases. In modal abundance, olivine amounts to 70–90%; pyroxenes (mostly orthopyroxene) to 1–8%; and plagioclase, commonly to <1%. About 1–2% chromite is generally present; and about 1–3% magnetite occurs, mostly associated with the sulfides. Olivine occurs as medium-grained (0.5–6 mm) euhedral to subhedral crystals. It is everywhere intensely altered to serpentine and magnetite, and, as seen under the microscope, it becomes opaque when it is intensely altered and replaced by magnetite (Fig. 5a). In extreme cases, where all of the olivine is serpentinized, only the outlines of its crystals remain discernible. Orthopyroxene typically occurs as reaction rings around olivine, and it is sometimes partly reacted to clinopyroxene. Chromite grains are invariably small (0.01–0.02 mm) and euhedral. They mostly occur as inclusions in olivine, but some are interstitial to the latter. These textures are typical of olivine chromite-cumulates in many intrusions and indicate that the chromite crystallized either before or at the same time as olivine. The sulfides in dunite are also interstitial; they typically occur in the middle of the interstitial voids, with interstitial pyroxene and plagioclase separating them from the cumulus olivine (Fig. 5b). In extreme cases, dunite consists entirely of olivine and interstitial sulfides, but in these circumstances, the olivine occurs as equant grains with slightly rounded crystal forms, as though it had undergone some resorption before the sulfides formed.

Lherzolite is by far the dominant rock type of the Jinchuan intrusion, making up ~70% of its total volume (S.G.U., 1984). It is generally massive with no visible layering, and it is essentially an olivine–chromite mesocumulate with interstitial pyroxenes, sulfides, and a small amount of plagioclase. Olivine comprises 35–85% of the mode; the pyroxenes, 10–50%. Orthopyroxene is predominant where olivine is abundant; clinopyroxene, where

olivine is less common. Plagioclase generally occurs in amounts $<2\%$, and cumulus chromite ranges from 0.5 to 2%. The grain size of olivine is highly variable, ranging from 0.2 to 8 mm. In the western part of the intrusion, olivine in the dunite core and the lower part of the intrusion is generally coarse grained with crystal size ranging from 2 to 8 mm, whereas the crystals of olivine in the marginal olivine pyroxenite are typically very fine grained, ranging from 0.3 to 3 mm. In the eastern subchamber, the grain size of the olivine ranges from 0.5 to 4 mm, gradually increasing from the base upward and decreasing from the centre of the intrusion outward to its margins. Olivine in lherzolite is invariably euhedral to

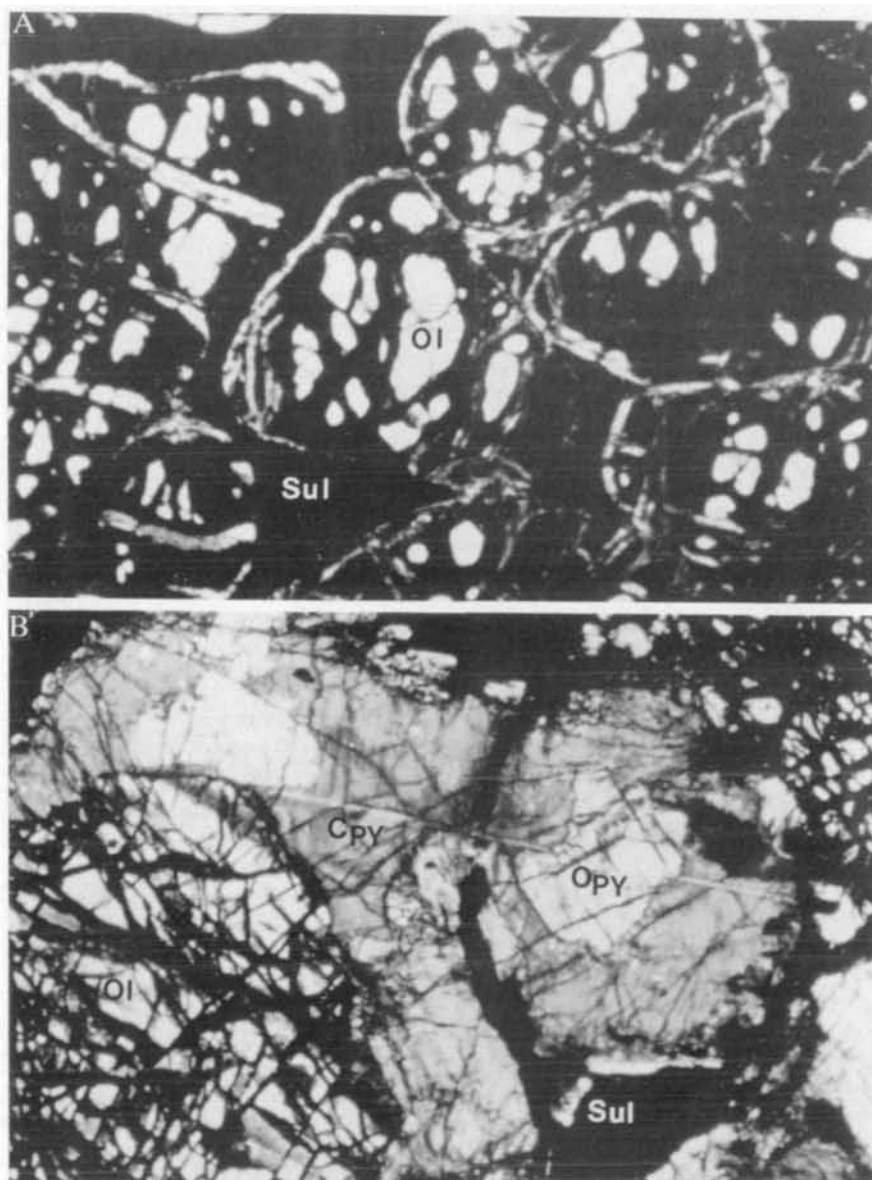


FIG. 5(A-B)

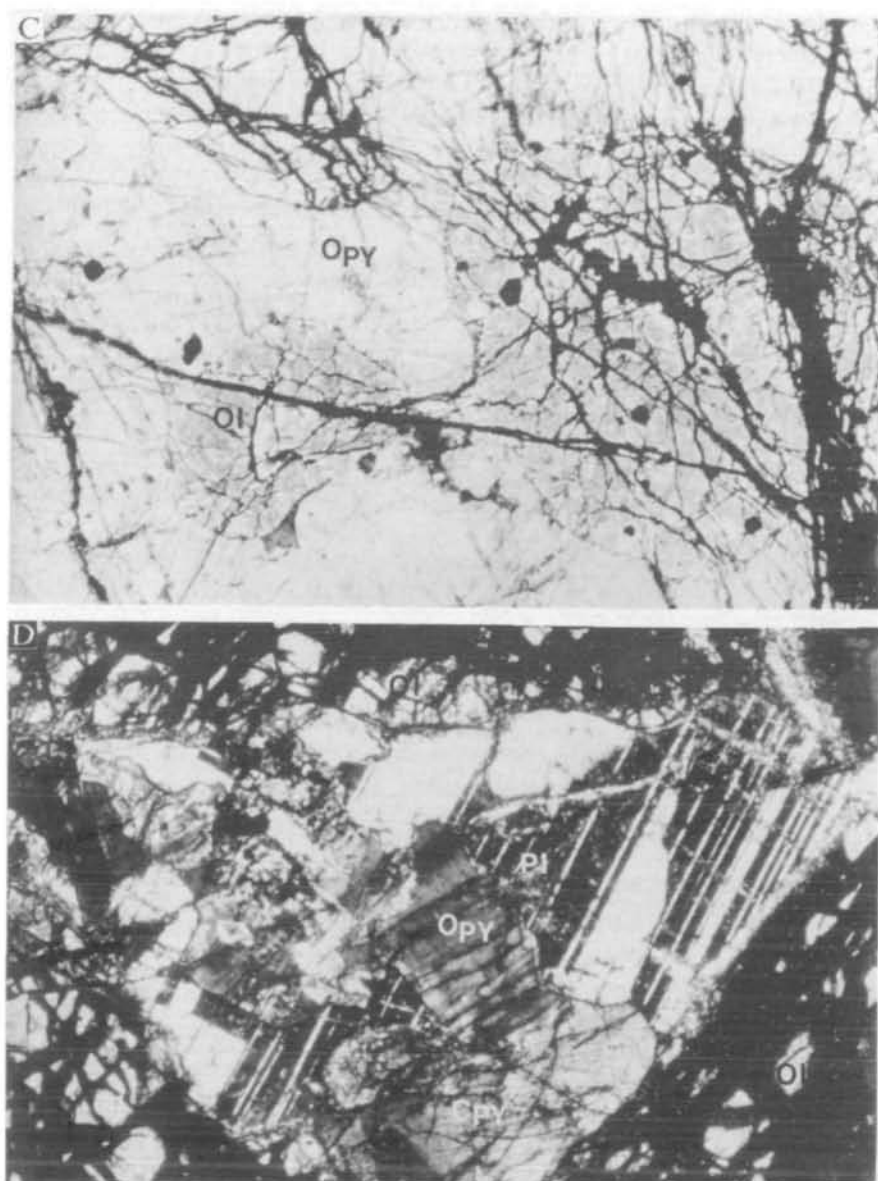


FIG. 5. Photomicrographs of rocks from the Jinchuan intrusion: (a) sulfide-bearing dunite; (b) lherzolite in which olivine is partly replaced by orthopyroxene, which is in turn replaced by clinopyroxene; (c) poikilitic texture showing olivine grains enclosed in orthopyroxene; (d) plagioclase lherzolite in which plagioclase and pyroxene are interstitial to cumulus olivine. In all cases, the field of view is 2.5 mm wide.

subhedral. The coarse-grained olivine commonly has irregular margins because of replacement by pyroxene and amphibole, and in altered rocks, it is locally cross-cut by talc. Medium-grained olivine commonly has rounded corners and is frequently enclosed by pyroxene oikocrysts (Fig. 5c). In lherzolite, olivine is generally less serpentinized, and fresh olivine grains are common, especially in pyroxene oikocrysts. The pyroxenes are characteristically interstitial to cumulus olivine and are frequently replaced by amphibole, chlorite,

and serpentine. They may occur as reaction rims around olivine, or as oikocrysts enclosing cumulus olivine grains. Where pyroxene forms reaction rims on olivine, sulfide is sometimes present, filling the gaps between the olivine–pyroxene aggregates. Where the pyroxenes are severely altered, it is difficult to distinguish orthopyroxene and clinopyroxene, but in other circumstances, clinopyroxene can commonly be seen either enclosing orthopyroxene or as an independent interstitial phase. There is generally <8% sulfide in the lherzolite, and the amount decreases upward. The upper part of the intrusion in the east is practically barren of sulfides, but mineralization extends to the surface in the west (Fig. 3.1).

Plagioclase lherzolite is described separately because of its importance as a visible marker layer in the eastern part of the intrusion. It generally contains 3–10% of plagioclase that typically is an interstitial phase along with pyroxene (Fig. 5d) and that commonly is sausseritized or chloritized. An average modal composition for plagioclase lherzolite is 74% olivine, 17% pyroxene, and 6% plagioclase. Small amounts of biotite and magnetite are also present. Sulfides are either absent or present in only trace amounts.

Olivine pyroxenite makes up <10% of the intrusion. It contains 10–40% olivine, and the rest of the rock is mostly clinopyroxene. Olivine pyroxenite is much fresher than the other rock types, and the olivine is generally unaltered. Pyroxene may be slightly altered to chlorite and amphibole. One pyroxenitic sample, collected at the contact between the intrusion and biotite gneiss, contains mainly pyroxene altered to amphibole and a small amount of biotite, and it may reflect some degree of contamination by the country rock. The olivine pyroxenite is generally mesocumulate or orthocumulate with pyroxene oikocrysts enclosing olivine. The olivine is much finer grained (0.2–2 mm) than that in other rock types. Traces of sulfides are present in some samples.

CHEMICAL COMPOSITIONS

Sample analytical procedure

A total of 60 whole-rock samples from the intrusion, mainly from the three drill holes represented in Fig. 4, have been analyzed for major elements by X-ray fluorescence (XRF) at the X-ray Assay Laboratory, Ontario, Canada. Trace elements (Cr, Rb, Sr, Y, Zr, Nb, Ba, and V) were analyzed by XRF in the Department of Geology, University of Toronto.

Most (54) of the above samples were also analyzed for rare earth elements (REE) and for Th, U, Sc, and Hf by instrumental neutron activation analysis (INAA) in the SLOWPOKE reactor, University of Toronto. Rock powders of between 450 and 550 mg were used for each sample, and the irradiated powders were counted for >6 h on the Canberra detector to obtain satisfactory results from the ultramafic rocks, which are low in REE.

Major elements

Representative major element analyses of the Jinchuan intrusive rocks are presented in Table 1. They have been recalculated to an anhydrous basis, and sulfide contents have been subtracted by the monosulfide calculation method of Naldrett (1981). All the rocks are high in MgO (24–42 wt.%). Dunite contains 39–42 wt.% MgO and is very low in Al₂O₃, CaO, TiO₂, Na₂O, and K₂O, all of which approach zero in the most magnesian samples. Lherzolite contains 26–39 wt.% MgO, with most samples having between 30 and 35 wt.%. Olivine pyroxenite commonly has about 25 wt.% MgO. The abundances of SiO₂, Al₂O₃, CaO, and TiO₂ all increase gradually from dunite to olivine pyroxenite. An average value of 30 wt.% MgO has previously been reported on the Jinchuan intrusion and was taken as the

composition of the primary magma (S.G.U., 1984). This value is about the generally accepted upper MgO content for komatiites and their related magmas (Naldrett, 1981; Barnes *et al.*, 1988b).

In Fig. 6, the major elements are plotted against MgO content. In these plots, samples with > 1 wt.% sulfur are omitted to reduce the effect of the incomplete subtraction of iron in sulfide by the monosulfide method. Systematic trends can be seen for most of the elements. In particular, SiO₂, Al₂O₃, CaO, and TiO₂ show strong negative correlations with MgO. The least-squares regression lines for SiO₂, Al₂O₃, and CaO have *R* factors of > 0.9; the value for TiO₂ is 0.86. The lines of Al₂O₃, CaO, and TiO₂ all intercept the ordinate at about 42–46 wt.% MgO, equivalent to ~38 wt.% SiO₂, roughly the amount in pure olivine in the Jinchuan rocks. However, the intercepts for Al₂O₃ and TiO₂ are slightly higher, possibly because they are also controlled by chromite and by ilmenite, which is seen to occur along chromite cleavages. Thus, at an MgO content of 44 wt.%, roughly that of pure olivine,

TABLE 1
Chemical composition of the Jinchuan intrusive rocks

Sample no. Rock type	I-13 Dun	I-17 Dun	I-18 Dun	II-26 Dun	I-4 Lz	I-6 Lz	I-8 Lz	II-7 Lz
wt. %								
SiO ₂	39.76	39.70	41.35	38.86	41.00	41.22	41.03	42.50
TiO ₂	0.16	0.12	0.19	0.13	0.29	0.31	0.28	0.38
Al ₂ O ₃	1.34	1.08	1.79	1.48	2.75	2.29	2.25	2.79
Cr ₂ O ₃	0.67	0.74	0.71	0.44	0.76	0.70	0.75	0.68
FeO	15.56	15.08	14.64	17.34	14.58	14.50	14.60	15.05
MnO	0.21	0.22	0.18	0.27	0.21	0.22	0.22	0.20
MgO	41.17	42.52	39.66	39.99	38.15	38.73	38.97	35.95
CaO	0.46	0.20	1.19	1.33	1.64	1.42	1.40	2.05
Na ₂ O	0.06	0.07	0.10	0.06	0.23	0.25	0.23	0.13
K ₂ O	0.32	0.06	0.09	0.03	0.27	0.16	0.16	0.15
P ₂ O ₅	0.04	0.04	0.03	0.03	0.04	0.06	0.05	0.05
Sum	99.08	99.09	99.22	99.51	99.17	99.14	99.18	99.26
mg-no.	0.83	0.83	0.83	0.80	0.82	0.83	0.83	0.81
ppm								
Ni	2653	4716	4269	5227	2143	1638	2005	1572
Co	197		213	243	176	154	139	146
Cr	4580	5041	4843	3009	5172	4807	5129	4640
V	44.7	44.5	52.6	49.4	75.3	69.7	73.5	91.7
Nb	2.3	1.6	2.9	2.1	3.5	3.2	2.5	3.6
Zr	11.1	7.5	16.8	5.1	24.0	23.8	19.8	36.7
Y	1.6	0.1	2.3	0.5	4.1	3.3	3.9	6.7
Sr	1.2	3.7	12.3	28.7	21.7	39.1	31.3	76.8
Rb	17.9	3.0	4.7	0.9	9.6	6.6	6.2	7.8
Cs	0.52		0.23	0.36	1.02	0.58	0.49	1.92
U	0.00		0.14	0.05	0.03	0.09	0.06	0.06
Th	0.42		1.21	0.00	0.57	0.34	0.25	0.38
Sc	7.1		5.4	7.1	8.9	9.0	7.8	13.7
Hf	0.32		0.40	0.00	0.61	0.57	0.43	0.74
La	0.78		0.64	0.88	3.08	2.66	2.71	5.10
Ce	2.06		2.62	1.80	7.46	5.54	5.63	10.79
Nd	1.19		2.08	0.88	3.32	2.80	2.16	5.10
Sm	0.45		0.56	0.20	0.88	0.86	0.68	1.18
Eu	0.06		0.09	0.06	0.29	0.24	0.19	0.36
Tb	0.06		0.08	0.03	0.11	0.13	0.10	0.19
Yb	0.22		0.21	0.06	0.40	0.36	0.35	0.55
Lu	0.04		0.04	0.02	0.06	0.06	0.06	0.08

TABLE 1 (Continued)

Sample no. Rock type	II-10 Lz	II-12 Lz	II-15 Lz	II-w-16 Lz	I-25 Ol Px	I-29 Ol Px	II-w-7 Ol Px	II-w-10 Ol Px
wt. %								
SiO ₂	41.70	42.07	40.42	42.56	46.21	43.98	45.71	45.62
TiO ₂	0.33	0.40	0.41	0.21	0.30	0.26	0.52	0.52
Al ₂ O ₃	2.74	2.76	2.16	4.00	4.93	4.59	6.81	6.72
Cr ₂ O ₃	0.78	0.64	0.67	0.57	0.60	0.85	0.50	0.50
FeO	15.11	14.79	16.49	15.45	13.67	16.17	13.21	13.21
MnO	0.23	0.22	0.22	0.21	0.26	0.28	0.18	0.18
MgO	37.16	36.76	36.33	33.75	25.73	25.59	25.70	25.81
CaO	1.75	1.86	1.81	2.31	7.69	6.30	5.93	5.95
Na ₂ O	0.09	0.15	0.26	0.29	0.11	0.04	0.87	0.91
K ₂ O	0.03	0.22	0.13	0.51	0.04	0.02	0.50	0.49
P ₂ O ₅	0.06	0.07	0.08	0.03	0.04	0.04	0.06	0.06
Sum	99.20	99.29	98.30	99.33	98.97	97.27	99.49	99.46
mg no.	0.81	0.82	0.80	0.80	0.77	0.74	0.78	0.78
ppm								
Ni	1613	1604	3920	5268	4641	4688	877	922
Co	168	165	286	251	301	242	88	96
Cr	5323	4408	4605	3866	4118	5803	3452	3410
V	86.6	90.6	86.9	74.8	136.6	126.1	140.4	136.1
Nb	2.3	3.5	4.6	2.7	2.6	1.5	4.4	4.6
Zr	27.4	37.6	45.9	19.1	17.3	26.0	48.3	47.3
Y	4.2	5.1	6.3	4.7	3.9	3.8	10.5	10.3
Sr	53.2	49.7	41.0	56.4	76.0	81.4	90.7	92.4
Rb	3.3	10.3	5.7	22.5	2.4	2.0	20.6	20.3
Cs	0.44	0.75	0.45	2.06	0.46	0.10	1.96	1.77
U	0.08	0.06	0.10	0.04	0.10	0.34	0.17	0.20
Th	0.35	0.51	0.49	0.33	0.51	2.55	0.75	0.78
Sc	12.7	12.7	11.2	12.2	21.0	11.9	20.2	17.4
Hf	0.69	0.95	0.99	0.40	0.52	0.70	1.43	1.27
La	4.02	4.96	3.66	2.15	1.29	20.59	3.72	3.94
Ce	9.09	10.20	8.06	5.57	3.21	29.03	12.84	12.58
Nd	4.34	4.91	5.40	3.36	2.40	9.26	5.86	7.28
Sm	1.13	1.20	1.26	0.68	0.66	2.03	1.68	1.67
Eu	0.38	0.38	0.35	0.28	0.21	0.31	0.49	0.57
Tb	0.19	0.15	0.19	0.16	0.13	0.13	0.28	0.24
Yb	0.55	0.61	0.59	0.41	0.55	0.46	0.79	0.86
Lu	0.10	0.12	0.12	0.08	0.09	0.06	0.12	0.14

mg-no., Molecular ratio of MgO/(MgO+FeO); Dun, dunite; Lz, Lherzolite; Ol Px, olivine pyroxenite.

Al₂O₃ and TiO₂ are roughly 0.5 and 0.1 wt.% respectively, about the maximal amounts in chromite. Another possibility is that there was some loss of CaO during alteration, and that this led to its lower intercept for MgO. The linear trends of these elements against MgO would appear, therefore, to be a mixing line between cumulus olivine and a magmatic liquid that is the true parental magma of the Jinchuan intrusion.

Generally, FeO increases with increasing MgO, ranging from 18 wt.% in the most magnesian rocks to 12 wt. % in those with the least MgO. The correlation, however, is not as strong as those of SiO₂, Al₂O₃, CaO, and TiO₂, and the scatter shown by the more magnesian rocks correlates closely with the presence of sulfides, possibly because the monosulfide calculation is not accurate enough to subtract sulfide iron completely, or because correlations have not been made for the magnetite and chromite present in the samples. Values of Cr₂O₃ typically are between 0.6 and 2 wt.%, and as olivine can only contain very limited amounts (<0.24 wt.%), most of this Cr₂O₃ in the rock can be ascribed

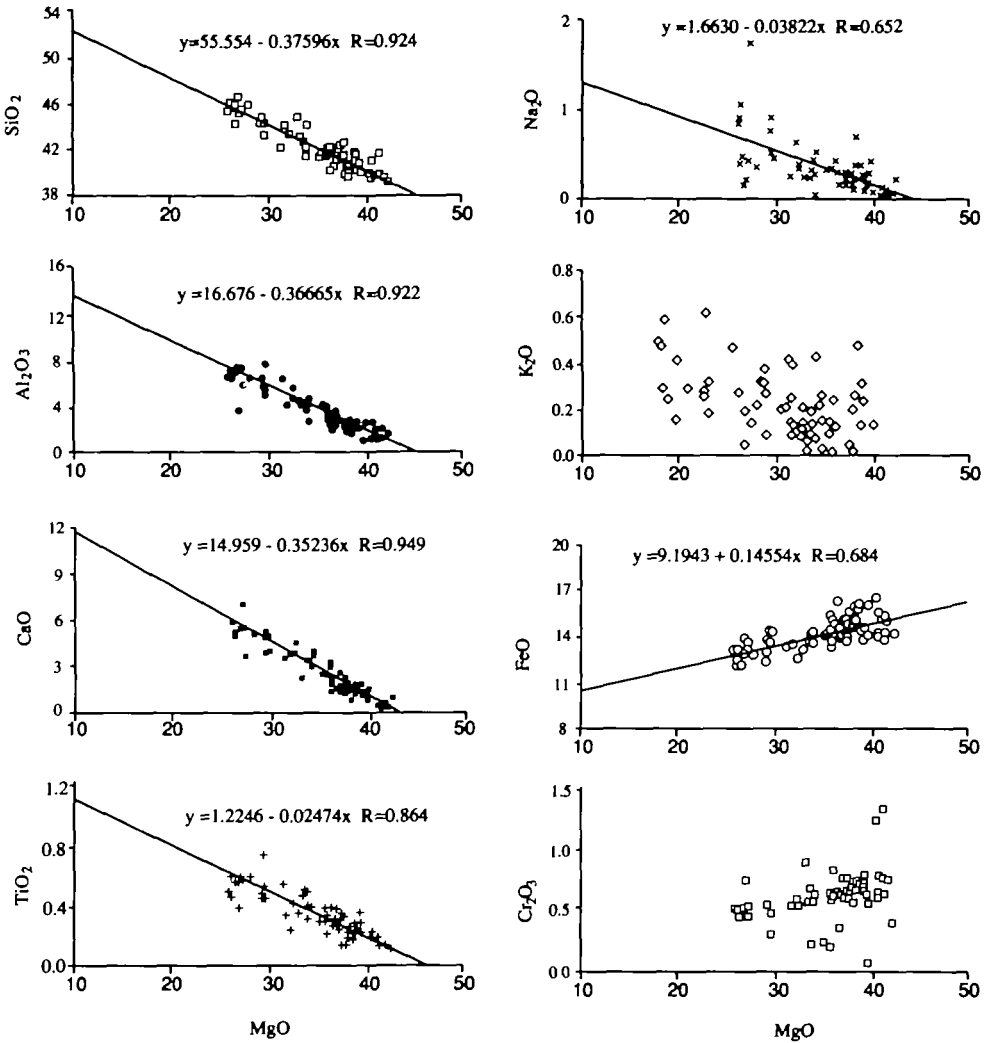


FIG. 6. Plots of whole-rock variation of major elements with MgO. The plots of SiO₂, Al₂O₃, CaO, and TiO₂ all show strong linear negative correlations with MgO, whereas FeO increases with MgO, and the other elements are relatively scattered. All elements are presented in weight per cent.

to chromite. The rough increase of Cr₂O₃ with MgO (Fig. 6h) probably reflects cotectic crystallization of olivine and chromite.

The plots of Na₂O and K₂O against MgO, especially the latter, show even more scatter than the other oxides. There is a very weak decreasing trend of Na₂O with MgO, but K₂O does not show any visible relationship. If the Jinchuan rocks can be regarded as mixtures of olivine and magmatic liquid, then because olivine contains little or no Na₂O and K₂O, their concentrations should primarily reflect the amount and composition of the interstitial liquid. Their scatter, therefore, is probably not the result of magmatic fractionation but is more likely to be an effect of the hydrothermal alteration that is ubiquitous in the intrusion.

The possibility of mixing of olivine and the magmatic liquid can be best demonstrated on the plots of molecular (MgO + FeO)/Al₂O₃ vs. SiO₂/Al₂O₃ and SiO₂/TiO₂ (Fig. 7a, b). The

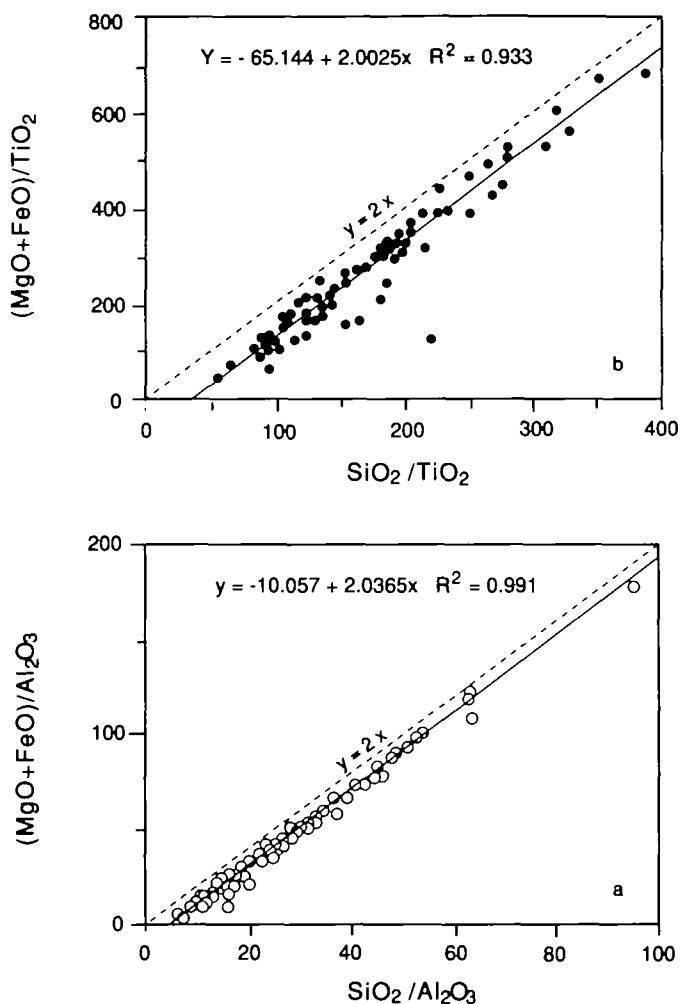


FIG. 7. Pearce molecular plots of the ratios of $(\text{MgO} + \text{FeO})$ and SiO_2 to (a) TiO_2 and (b) Al_2O_3 . The slope of $(\text{MgO} + \text{FeO})/\text{SiO}_2$ is close to two, which is the ratio of olivine, indicating olivine-controlled fractionation.

regression lines in these plots have R^2 factors of >0.93 , and both have slopes within the range of 2 ± 0.05 , essentially the molecular value of $(\text{MgO} + \text{FeO})/\text{SiO}_2$ in olivine. Thus, it is clear that the composition variations of the Jinchuan intrusive rocks are dominantly controlled by olivine. The regression lines do not pass through the origin as would ideally be expected (Pearce, 1968), but if 0.1 wt.% of TiO_2 and 0.5 wt.% of Al_2O_3 are subtracted, corresponding approximately to the amounts in the cumulus chromite, then the lines do meet the origin.

Trace elements

Selected trace element analyses of the Jinchuan rocks are also listed in Table 1. The values for Rb and Ba contents are very low, close to their detection limit, and those for Ni commonly reflect the presence of sulfides. All other trace elements are low in dunite and

increase through lherzolite to olivine pyroxenite. The most impressive correlations are the linear covariations of the trace elements with TiO_2 ; most are positive, particularly those of the hydrothermally immobile elements such as Hf, Zr, V, and Y (Fig. 8). The correlation of Nb with TiO_2 is not as linear as those of the other trace elements, possibly because the amounts of Nb (1–7 ppm) are close to the detection limit of the analytical method. The data

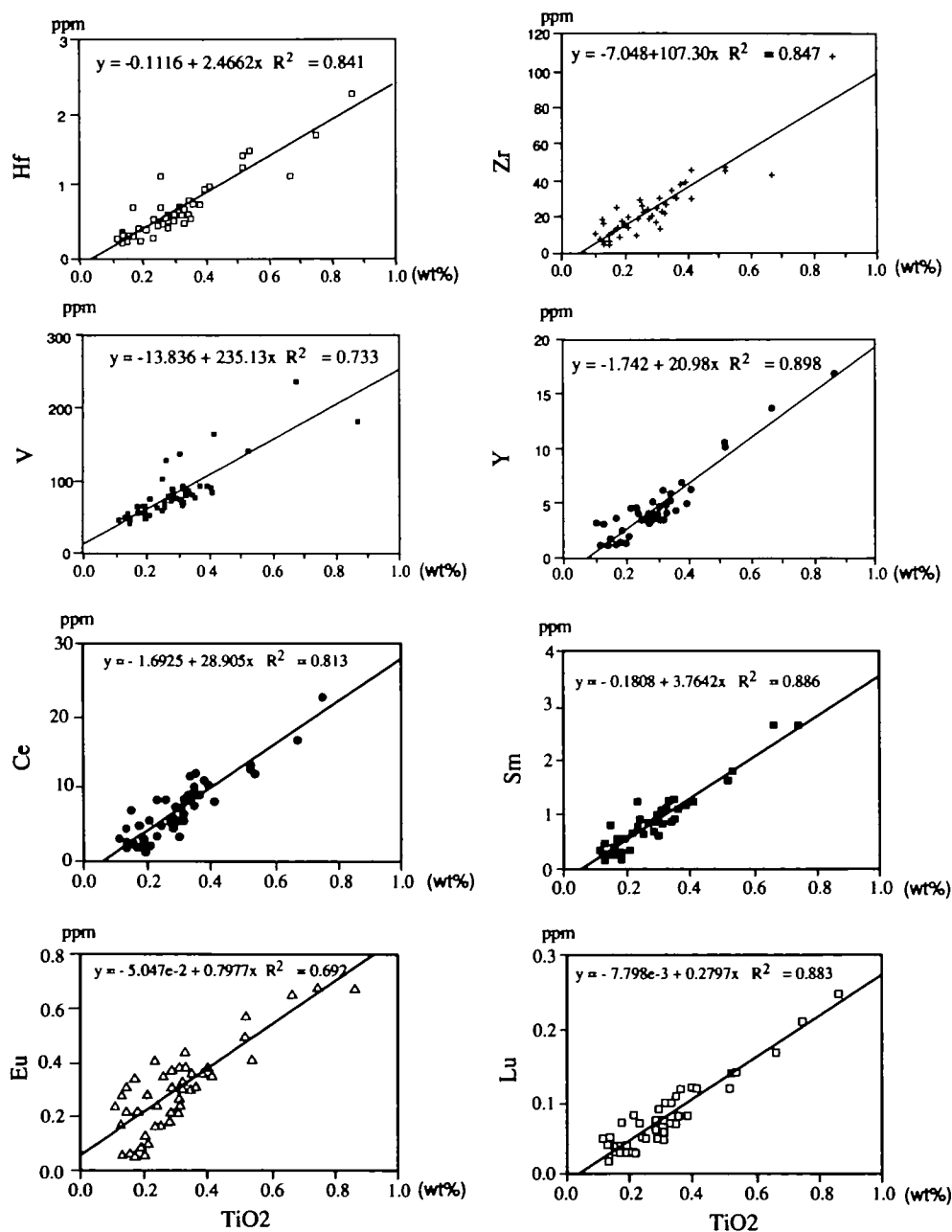


FIG. 8. The relationship of selected trace elements and REE with TiO_2 . They all show positive correlation with TiO_2 .

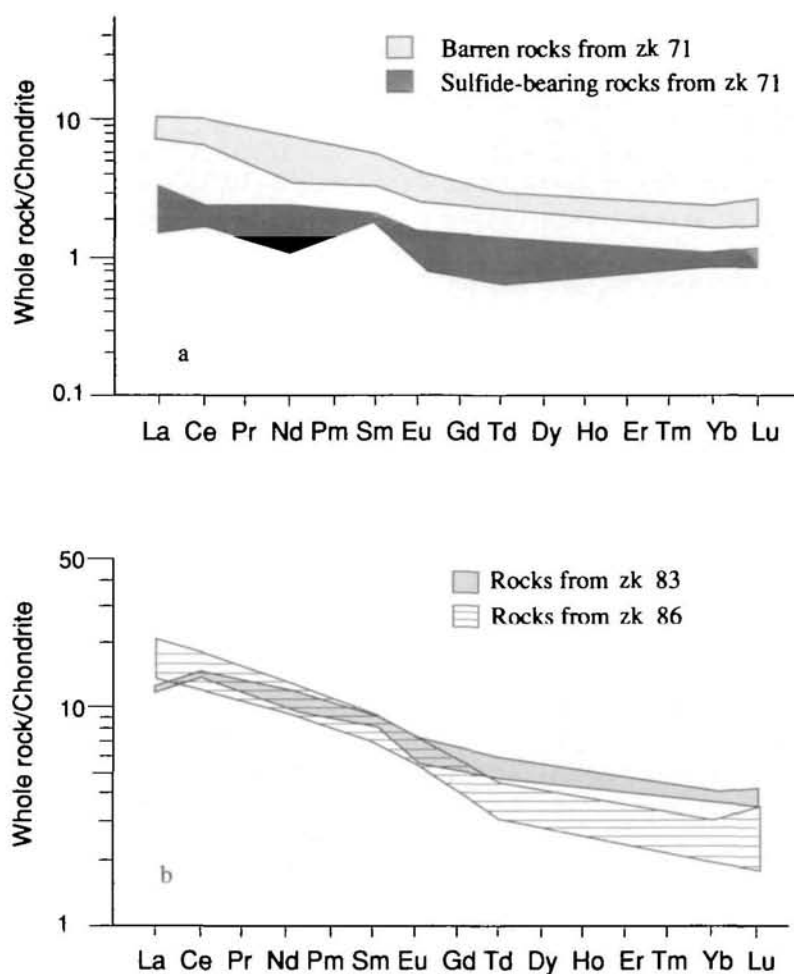


FIG. 9. Chondrite-normalized REE patterns for rock samples from drill holes zk71 (a), and zk83 and zk86 (b). The barren rock samples have very similar REE patterns and similar range (each envelope represents > 10 samples). Sulfide-bearing samples have lower REE concentrations.

for Sr are also scattered, and the likely reason is its mobility during hydrothermal alteration, which is similar to those of Na_2O and K_2O (Ludden *et al.*, 1983). Inasmuch as TiO_2 is incompatible in the Jinchuan rocks, then those trace elements that are positively correlated with TiO_2 must also be incompatible.

Rare earth elements (REE)

Chondrite-normalized REE patterns of the Jinchuan rocks are shown in Fig. 9 and representative REE analyses are also listed in Table 1. Samples from drill hole zk71 include both the barren and sulfide-bearing rocks.

The rocks all show patterns of LREE enrichment. Thus, in the sulfide-free rocks, the LREE values are generally ~8–20 times the chondrite abundance, whereas the HREE values are only 2–6 times as large. The variations from La to Lu are relatively smooth with no obvious Eu anomalies; the chondrite-normalized La/Lu ratios are between 4 and 10. The

tendency to LREE enrichment is slightly stronger in drill hole zk86 from the eastern subchamber of the intrusion (Fig. 3.2d), where MgO contents are generally lower (around 30–37 wt.%). These features suggest that the magma in this subchamber was somewhat more evolved.

The REE patterns of the sulfide-rich samples from drill hole zk71 are broadly similar to the patterns of other rocks, except that concentrations are 2–5 times less, and the La/Lu ratio is slightly lower (2–5). In the sulfide-bearing rocks, the patterns are slightly irregular because the REE concentrations are so low that they are almost at the detection limit; the higher sulfide content also increases the background, making it more difficult to define the peaks of these elements during the analysis. Because the sulfides occur interstitially in the olivine cumulates, the REE abundances essentially reflect the composition and abundance of the silicate interstitial liquid. The near-parallel REE distribution patterns for the Jinchuan rocks also reflect an olivine-controlled differentiation process, because olivine accommodates almost no REE (Cox *et al.*, 1979).

The REE contents of the rocks invariably show positive linear correlations with TiO₂, as is illustrated by the plots of four examples in Fig. 8. This feature has provided further support for the olivine-controlled crystallization process of the Jinchuan rocks.

Olivine compositions

Olivine in the Jinchuan rocks was analyzed by a fully automated Cameca-SX50 electron microprobe in the Department of Geology, University of Toronto. We used an accelerating voltage of 15 kV, a beam current of 20 nA, and a beam diameter of 2 µm. Synthetic olivines, ranging from forsterite to fayalite, were used as standards for Si, Mg, and Fe, with 15 s counting time for each element; metal Ni was used for Ni, with 50 s of counting time. Each olivine composition is an average of 4–7 analyses. The analytical results are presented in Table 2.

Forsterite contents (mol. % Fo) of olivine in the Jinchuan rocks are in the range of 84–86% in dunite, 82–85% in lherzolite, and 79–82% in olivine pyroxenite. Little within-sample variation is observed for olivine (<1%), which contains 43–44.5 wt.% MgO, in accordance with the values of 42–46 wt.% MgO extrapolated from the whole-rock data.

TABLE 2
Chemical composition of olivine

Sample no.	SiO ₂ (wt.%)	MgO (wt.%)	FeO (wt.%)	NiO (wt.%)	Total	Fo	mg-no. of liquid
II-w-8	39.97	40.92	18.11	0.18	99.19	80.11	54.73
II-w-9	40.70	43.63	14.98	0.29	99.60	84	61
II-w-12	40.99	44.38	14.69	0.25	100.31	84.33	61.76
II-w-16	40.83	43.67	15.39	0.28	100.17	83.49	60.28
II-w-19	40.90	44.33	14.15	0.27	99.65	84.82	62.63
II-w-20	40.88	44.18	14.76	0.27	100.08	84.22	61.55
II-w-21	40.73	44.37	13.50	0.15	98.74	85.52	63.82
2-E-9	40.81	43.00	15.85	0.21	99.87	82.87	59.20
2-w-10	40.74	44.34	13.91	0.20	99.18	85.04	63.03
2-w-14	40.95	43.94	14.57	0.22	99.68	84.31	61.72
O-4	40.84	43.92	15.55	0.24	100.04	83.43	60.17
I-4	40.92	44.13	14.76	0.21	100.02	84.20	61.51
I-5	40.92	44.12	14.75	0.24	100.02	84.21	61.54
I-9	41.00	44.20	14.42	0.22	99.84	84.53	62.11
I-12	41.06	44.18	14.23	0.25	99.73	84.69	62.40

Fo, Forsterite content; mg-no., mole ratio of MgO/(MgO + FeO).

Abundances of NiO in olivine range from 0.15 to 0.3 wt.% and generally increase slightly with increasing Fo content. No significant difference was observed between mineralized and unmineralized rocks, although most of the olivine in the mineralized rocks has higher MgO content (~ 44 wt.%, Fig. 10).

PETROGENETIC CONSIDERATIONS

Nature of the primary magma

MgO/(MgO + FeO) of primary magma

Except for one sample from the contact between the intrusion and biotite gneiss, consisting of amphibole and biotite, all the intrusive rocks appear texturally to be olivine–chromite cumulates. This observation is also confirmed by the geochemical data, but no chilled margin has been observed in the intrusion, and no closely related mafic dykes have been identified as possible feeders; hence there is no direct evidence of the composition of the parental magma. This composition must, therefore, be inferred in some indirect way.

One approach is to use the olivine composition to estimate the MgO/FeO value for the magmatic liquid by way of the Roeder & Emslie (1970) Mg–Fe distribution coefficient:

$$K_D = (\text{FeO}/\text{MgO})^{\text{Ol}} / (\text{FeO}/\text{MgO})^{\text{L}} = 0.3 \pm 0.03.$$

This coefficient is relatively insensitive to temperature, pressure, and liquid composition (Roeder & Emslie, 1970). One complication is that cumulus olivine is usually less magnesian than it was during its original crystallization, because of re-equilibration with the inter-cumulus liquid, so the calculated Mg/(Mg + Fe) value may be lower than that of the primary magma (Barnes, 1985; Chalokwu & Grant, 1987; Paktunc, 1987, 1989) and an adjustment may have to be made (Irvine, 1974). However, two features of the Jinchuan olivine indicate that it may be very close to its original composition. First, it does not appear to have

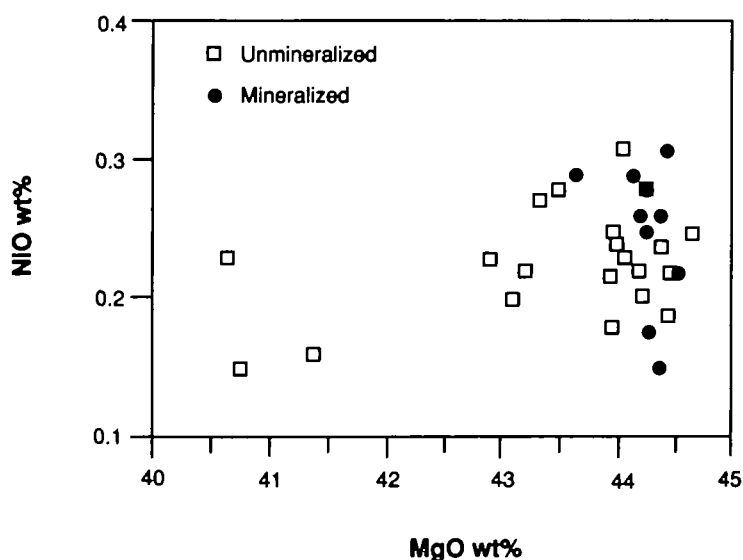


FIG. 10. Plot of NiO against MgO for olivine from the Jinchuan intrusion. (Note that no difference can be seen between olivines from sulfide-bearing and barren rocks.)

undergone much overgrowth; in fact, it is more commonly resorbed by pyroxene. And second, olivines immersed in sulfides may not always have been in contact with interstitial silicate liquid; thus olivine may be able to keep its original composition. The contact of olivine with sulfide may have some effect on the Fe content of olivine during subsolidus re-equilibrium, but this effect is usually negligible (Rajamani & Naldrett, 1978). The $\text{MgO}/(\text{MgO} + \text{FeO})$ values calculated for the liquids equilibrated with olivine are presented in Table 3. The most magnesian olivine in the intrusion is $\sim \text{Fo}_{86}$, which corresponds to an $\text{MgO}/(\text{MgO} + \text{FeO})$ value of liquid of 0.64, based on a distribution coefficient of 0.3. The latter value is much lower than the $\text{MgO}/(\text{MgO} + \text{FeO})$ ratios of any of the Jinchuan intrusive rocks, so none of them can represent the parental magma, nor an average composition for the intrusion.

Composition of the primary magma

To estimate the composition of the Jinchuan parental magma, more information is required other than the $\text{MgO}/(\text{MgO} + \text{FeO})$ value. One method of estimation uses $\text{MgO}/(\text{MgO} + \text{FeO})$ values calculated from olivine compositions to compare with a chilled margin composition (Irvine, 1980; Barnes, 1985; Duke, 1986). Other workers have attempted a mass balance method, using TiO_2 as an index of the proportion of the trapped liquid, together with a composition estimated from the spinifex textured komatiite (Barnes *et al.*, 1988b).

These methods cannot be applied to the Jinchuan intrusion, however, because it has no equivalent spinifex textures and no chilled margins. Previous researchers (S.G.U., 1984) considered the problem of initial liquid of the Jinchuan intrusion in connection with the lateral zonation of rock types and sulfide mineralization. They suggested that the intrusion is a composite formed from several injections of magmas compositionally equivalent to the observed rock types, and they calculated the parental magma composition simply as an average of the analyses of the different rock types weighted according to their volumes. The result showed 33.90 wt.% MgO and 14.53 wt.% FeO [with $\text{Fe}_2\text{O}_3/(\text{FeO} + \text{Fe}_2\text{O}_3) = 10\%$], but it is not satisfactory simply because the value of $\text{MgO}/(\text{MgO} + \text{FeO})$ is 0.81, whereas the appropriate value as calculated above is 0.64.

If, on the other hand, one ignores the minor chromite in the Jinchuan rocks, then they can be taken as mixtures of the primary magma and cumulus olivine. The negative linear correlations between the major elements and MgO (Fig. 6), and the positive linear correlation between the trace elements and REE and TiO_2 (Fig. 8) present strong support for this approach.

As we know only the $\text{MgO}/(\text{MgO} + \text{FeO})$ value (0.64) of the parental Jinchuan magma, a key to the problem is to define either MgO or FeO. To do so, we can first examine the relationship between FeO and MgO content of the rocks. If a trend of FeO vs. MgO can be defined, and if its slope is different from that defined by the $\text{MgO}/(\text{MgO} + \text{FeO})$ ratio, then the intercept of the regression line with the line representing an $\text{MgO}/(\text{MgO} + \text{FeO})$ value of 0.64 will give the MgO and FeO contents of the parental magma.

There are three problems in this approach: (1) the whole-rock samples must be sulfide free, because even a small amount of sulfide represents an addition of iron to the whole-rock composition that is difficult to correct precisely; (2) sulfide that has segregated from silicate magma characteristically contains a certain amount of oxide (Naldrett, 1969; Shima & Naldrett, 1975) that is also difficult to estimate; and (3) the original FeO values of the intrusive rocks are difficult to determine, because our XRF analyses give only total iron content. We have treated the total iron values as FeO, as the Jinchuan rocks are dominated

TABLE 3
Chemical composition of the Jinchuan primary magma and basalts from different tectonic settings

Element	Jinchuan magma	5% Crystal.	Low-Ti CRB*	High-Ti CRB*	N MORB†	P MORB†	CB‡	IACAB§	Kilauea‡
SiO ₂	50.8	51.33	51.52	51.37	50.55	49.72	46.18	49.4	50.51
Al ₂ O ₃	12.5	13.16	13.90	12.81	16.38	15.81	14.47	13.29	13.45
CaO	10.3	10.84	10.13	8.17	11.62	11.84	9.68	12.22	11.18
MgO	11.5	9.79	8.63	5.33	7.8	7.9	9.99	10.44	7.41
Fe ₂ O ₃	1.0	1.05	1.03	1.34	0.90	0.93	1.35	1.02	1.14
FeO	11.2	10.44	9.26	12.03	8.13	8.35	12.17	9.14	10.23
TiO ₂	1.0	1.05	1.14	3.38	1.15	1.46	2.06	0.7	2.63
Na ₂ O	1.3	1.37	2.27	2.56	2.43	2.35	2.63	2.16	2.28
K ₂ O	0.8	0.84	0.83	1.69	0.08	0.5	0.61	1.06	0.49
mg-no.	64.45	62.53	62.42	44.12	63.11	62.77	59.41	67.08	56.35
Hf	2	2.11					3.97		4.39
V	240	252.63	295		257	250			
Rb	30	31.58	24	33	1	9.57	14	14	9.2
Zr	100	105.26	120	275			167	40	115
Y	20	21.05	28	39			31	15	25
Ni	400		123	40	214	104	193		
Th	1.4	1.47					2	1.1	1.27
Sc	35	36.84			36	36.15			
Nb	8	8.42	11	29			15	1.4	17
Sr	200	210.53	189	455	106	243.6	285	550	357
Zr/Nb	12.50	13.16	10.91				11.06	28.57	6.76
Zr/Y	5.00	5.26	4.29	7.05			5.39	2.67	4.60
La	12	12.63	14.00	41.00	2.73	13.39	18.30	10	13.4
Ce	26	27.37	28.40	84.50			41.20	23	35.5
Nd	15	15.79	18.30	59.20			23.00	13	
Sm	3.5	3.68	3.13	10.07	3.72	3.93	5.60	2.9	6.14
Eu	0.9	0.95	0.94	3.38	1.12	1.3	1.94		1.88
Tb	0.4	0.42	0.53	1.75					
Yb	1.5	1.58	1.80	3.06	3.01	2.37	2.78		1.98
Lu	0.25	0.26	0.27	0.34			0.42		

Major elements are present in wt.%, trace elements and REE are in ppm. The third column is the composition of Jinchuan magma after 5% of olivine with 84% forsterite has crystallized. CRB, Continental rift basalt; N MORB, normal mid-ocean ridge basalt; P MORB, plume mid-ocean ridge basalt; CB, continental basalt; IACAB, island arc calc-alkaline basalt.

* Fodor *et al.* (1990). † Schilling *et al.* (1983). ‡ Basaltic Volcanism Study Project (1981). § Perfit *et al.* (1980).

by olivine, which contains almost no Fe_2O_3 . The small amounts of Fe_2O_3 contained in chromite and interstitial phases are ignored, but they may have some effect on the estimation.

For the above reasons, samples with $<0.2\%$ sulfur were selected and plotted on the FeO vs. MgO diagram (Fig. 11). It can be seen that they show a positive correlation with an R factor of 0.71. Furthermore, because the FeO trend is relatively flat, a slight variation of the regression line does not significantly change the intercepted MgO content. The intercept of the regression line at the $\text{MgO}/(\text{MgO} + \text{FeO})$ value line of the parental magma is at 11.5 wt.% FeO and 12 wt.% MgO. This result indicates that the parental magma was a high-MgO basalt compositionally far from an ultramafic magma with >30 wt.% MgO.

The amounts of other major elements in the Jinchuan parental magma were estimated from the MgO variation diagrams in Fig. 6, and the TiO_2 value obtained in this way was then used to estimate the abundance of some of the trace elements and all of the REE on the basis of Fig. 8. The liquid composition obtained in this way is listed in Table 3, together with comparative data from volcanic rocks in different tectonic settings.

The estimated Jinchuan magma composition differs in major element composition from mid-ocean ridge basalts (MORB) in that they have higher Al_2O_3 (16 wt.%); another difference is that normal MORB has an LREE-depleted pattern, whereas Jinchuan magma has an LREE-enriched pattern. The lower TiO_2 (1.0 wt.%) and Nb (8 ppm) contents of the Jinchuan magma exclude it from oceanic island tectonic setting. Rock compositions similar to that of the Jinchuan magma include the island arc calc-alkaline basalts and continental rift high-MgO basalts such as the Servilleta basalt of the Taos Plateau, New Mexico (Lipman & Mehnert, 1979) and the low-Ti basalt of Maranhão, Northern Brazil (Fodor *et al.*, 1990). Calc-alkaline basalts are slightly higher in CaO and lower in SiO_2 and TiO_2 than Jinchuan magma, and another typical difference is that they are much lower in Nb (1.4 ppm). On the other hand, if 5% of olivine with an Fo content of 84% is subtracted from the Jinchuan magma, the residual liquid is surprisingly similar to the high-MgO basalt of Maranhão. The SiO_2 , Al_2O_3 , MgO, FeO, and TiO_2 contents are almost identical, and trace

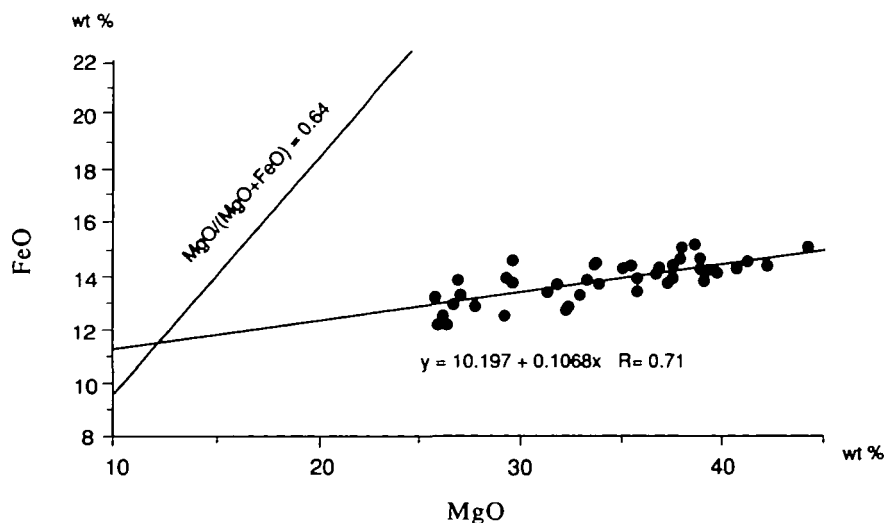


FIG. 11. Extrapolation of the variation diagram of FeO and MgO to estimate the composition of the Jinchuan primary magma. The primary magma has an MgO content of ~ 12 wt.%.

elements such as V, Zr, Y and Sr, and REE also match very well. Thus, the Jinchuan magma would appear to represent magmatism in a continental rift system.

The problem with this interpretation is the lack of evidence for rifting; the regional faults, F_1 and F_2 , define an uplifted region, not a graben. However, these faults may have been reactivated from originally tensional faults, a possibility that is supported by the presence of mafic-ultramafic intrusions along them. Another possibility is that a rift is hidden under the desert sands and has not been detected. A second problem with the rift model is the lack of mafic volcanic rocks. However, the region has undergone major uplift and erosion, and its edges are now concealed by thick Quaternary sand deposits. Thus, missing volcanic rocks could be either eroded away or deeply buried.

Emplacement and differentiation

The contrast between the lateral zonation in the western subchambers and the vertical layering in the eastern subchamber of the Jinchuan intrusion indicates that these subchambers were formed through different processes. In the western subchamber, the lateral zonation is marked by the change of rock types from dunite at the core, through lherzolite to olivine pyroxenite at the margin, and it is paralleled by a variation in the grain size of cumulus olivine from coarse grained at the centre to fine grained at the margin.

The grain size most probably reflects different cooling rates. The ratio of the depth to the width of the western subchambers is more than three, and there is no evidence of the top of the intrusion, thus lateral temperature gradients may have been much greater than the vertical gradient. The latent heat from crystallization in the magma may, therefore, have been largely conducted through the wall rocks. In such a case, the narrow marginal zone may have formed by rapid crystallization of stagnant, nearly chilled magma at the margins, and the coarse grain size and relatively high and constant forsterite content in the central zone rocks may indicate that the olivine there crystallized more slowly from magma of relatively constant composition.

The change of rock types associated with changes in the amount of cumulus olivine suggests that the differentiation process involved transverse movement of olivine crystals toward the central axis. This cannot be ascribed to crystal settling because that should produce vertical differentiation. But friction along the walls and differential cooling in the narrow conduit may have caused flow differentiation, a process whereby crystalline phases and denser liquid concentrate toward the centre of the conduit (Gibb, 1968; Barker, 1983); we favour this process for the differentiation of magma in the western subchambers of the Jinchuan intrusion.

In the case of the eastern subchamber, the cumulus texture and gross phase layering, as well as the widening of the magma chamber, suggest that it was formed dominantly by gravitational differentiation. The thinning of the lithological layers from the centre toward the margin implies that the floor of the magma chamber was inclined toward its centre, rather like that of the Great Dyke (Wilson & Prendergast, 1989), and the reappearance of lherzolite on top of plagioclase lherzolite can be attributed to a large addition of new magma.

If the western and eastern subchambers are compared, one sees that the western subchambers are deeper and narrower and contain coarser olivine; the eastern subchamber is relatively shallow and wide, has medium-grained olivine and shows a rough phase layering. Plagioclase lherzolite occurs only locally at the surface in the western part but is common in the eastern part, although it pinches out toward the west. These observations suggest that the eastern subchamber was at a higher part of the original magma chamber.

Furthermore, the slightly lower MgO content of the lherzolite in the eastern section (35–38 wt.%) (Fig. 4b) as compared with the western section (37–39 wt.%) (Fig. 4a), and the slightly greater degree of LREE enrichment of the rocks in the eastern section, also indicate a more fractionated magma in the eastern section. Because the Jinchuan intrusion has been unroofed by erosion and becomes wider upwards, the greater part of the original magma chamber was probably above the present level of exposure. Its general form is evidently dyke-like on the surface and trumpet-shaped in cross-section, so perhaps its form was similar to the Great Dyke in Zimbabwe (Wilson, 1982; Wilson & Prendergast, 1989) or the Skaergaard intrusion in Greenland (Wager & Brown, 1968).

Based on the above discussion, the formation of the Jinchuan intrusion is interpreted as follows (Fig. 12). A high-MgO basalt carrying olivine (chromite) crystals and, perhaps, sulfide droplets was intruded in the Jinchuan region. In the main conduit, which is thought to be represented by the two western subchambers, magma differentiation by a flow differentiation process that concentrated olivine crystals and sulfide droplets in the central part of the conduit. The continuous magma flow replenished the interstitial melt and resulted in overgrowth of olivine cumulate with constant composition. Towards the margin of this conduit, heat loss to the country rock caused more rapid crystallization of the magma and also increased its viscosity. In this way, a lateral zonation was formed, defined by outward decrease in both the amount and size of the olivine. In the upper and wider part of the magma chamber, which is shown by the eastern subchamber, magma spread horizontally where flow differentiation no longer took place. Differentiation here may have been controlled mainly by magma convection and gravitational differentiation (Huppert & Sparks, 1980; Irvine, 1980). Olivine started to accumulate as layers, and the liquid composition gradually evolved away from olivine. Repeated additions of new magma gave

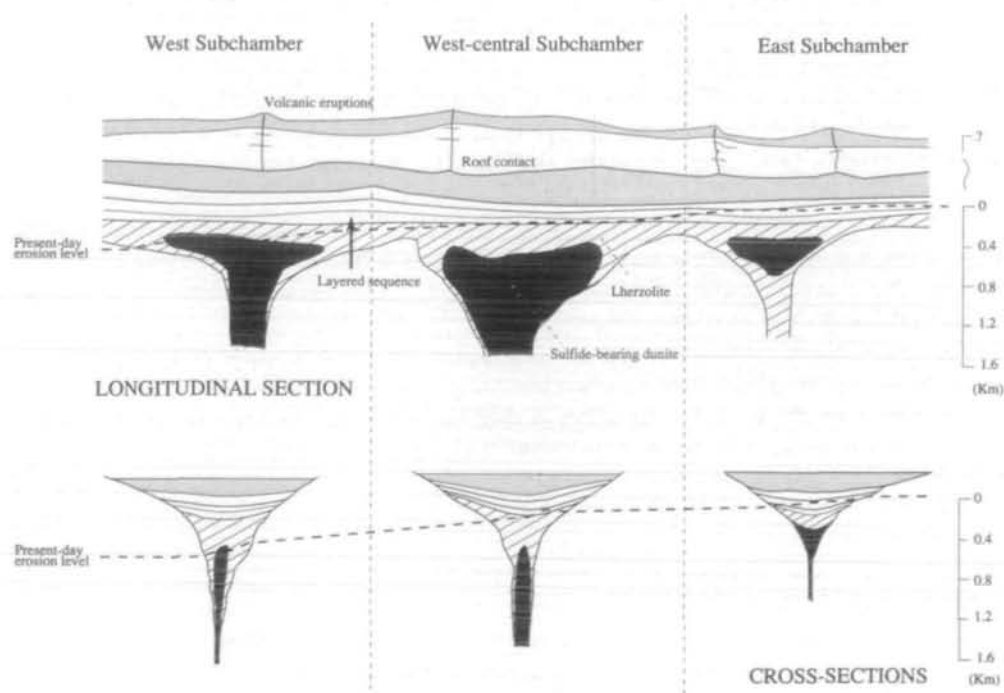


FIG. 12. Schematic sections illustrating our interpretation of the formation of the Jinchuan intrusion.

rise to the repetitive formation of olivine-bearing layers shown in Fig. 3.2d, and at higher levels in the magma chamber, a thick layered gabbroic sequence might have formed from the evolved magma. The magma chamber was probably open at the top to periodic loss of more evolved magma by volcanism. The absence of the layered ultramafic rocks in the western part, and of any mafic rocks anywhere in the intrusion suggests that we are only seeing a relatively deep section owing to the structural tilting to the east and greater erosion in the west.

CONCLUSIONS

From this study, the following conclusions can be reached:

- (1) The primary magma of the Jinchuan intrusion was not ultramafic as previously suggested, but was a high-MgO basalt with an MgO content of ~12 wt.%. The mafic rather than ultramafic nature of the magma is consistent with the Cu- and PGE-rich character of the sulfide ores.
- (2) The Jinchuan intrusive rocks are mainly olivine (chromite) cumulates. The upper part of the intrusion was probably gabbroic and some of the magma was probably erupted volcanically. The intrusion may have been a feeder for continental rift basaltic volcanism along a continental margin environment.
- (3) The western section of the intrusion was the main conduit to the magma chamber, the intrusion has been tilted eastward, and there has been more erosion in the west, exposing a deeper part of the body.
- (4) Most of the major and trace elements and all the REE were immobile during all the Jinchuan post-magmatic processes, but K_2O , Na_2O , and Rb were apparently affected by hydrothermal alteration, mainly serpentinization and chloritization.

ACKNOWLEDGEMENTS

We gratefully acknowledge Jinchuan Non-Ferrous Metal Co. for access to the mine and permission to collect samples. We thank S. Alfreyer, R.G.V. Hancock, and M.P. Gorton for help with the neutron activation analyses, and T. C. Barrie, C. De Ronde, and C. Li for reviewing a preliminary version of the manuscript. The first fieldwork by G. Chai was supported by a loan from the China University of Geosciences. Additional work was supported by NSERC operating grant A4244 to A. J. Naldrett and by a Graduate Research Award (88–89) from the Geological Society of America to G. Chai. The comments offered by three reviewers, and in particular, T. N. Irvine, have substantially improved the manuscript.

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